But the electric dipole moment $p = 2qa$ and we get,

$$V = \frac{1}{4\pi\varepsilon_0}\left(\frac{p\cos\theta}{r^2}\right)$$

Now we can write $p\cos\theta = \vec{p}\cdot\hat{r}$, where $\hat{r}$ is the unit vector from the point O to point P. Hence the electric potential at a point P due to an electric dipole is given by

$$V = \frac{1}{4\pi\varepsilon_\circ}\frac{\vec{p}\cdot\hat{r}}{r^2} \qquad (r>>a) \quad (1.38)$$

Equation (1.38) is valid for distances very large compared to the size of the dipole. But for a point dipole, the equation (1.38) is valid for any distance.

Special cases

Case (i) If the point P lies on the axial line of the dipole on the side of +q, then $\theta = 0$. Then the electric potential becomes

$$V = \frac{1}{4\pi\varepsilon_\circ}\frac{p}{r^2} \qquad (1.39)$$

Case (ii) If the point P lies on the axial line of the dipole on the side of –q, then $\theta = 180^\circ$, then

$$V = -\frac{1}{4\pi\varepsilon_\circ}\frac{p}{r^2} \qquad (1.40)$$

Case (iii) If the point P lies on the equatorial line of the dipole, then $\theta = 90^\circ$. Hence

$$V = 0 \qquad (1.41)$$

Important points

(i) The potential due to an electric dipole falls as $\frac{1}{r^2}$ and the potential due to a single point charge falls as $\frac{1}{r}$. Thus the potential due to the dipole falls faster than that due to a monopole (point charge). As the distance increases from electric dipole, the effects of positive and negative charges nullify each other.

(ii) The potential due to a point charge is spherically symmetric since it depends only on the distance r. But the potential due to a dipole is not spherically symmetric because the potential depends on the angle between $\vec{p}$ and position vector $\vec{r}$ of the point.
However the dipole potential is axially symmetric. If the position vector $\vec{r}$ is rotated about $\vec{p}$ by keeping θ fixed, then all points on the cone at the same distance r will have the same potential as shown in Figure 1.27. In this figure, all the points located on the blue curve will have the same potential.

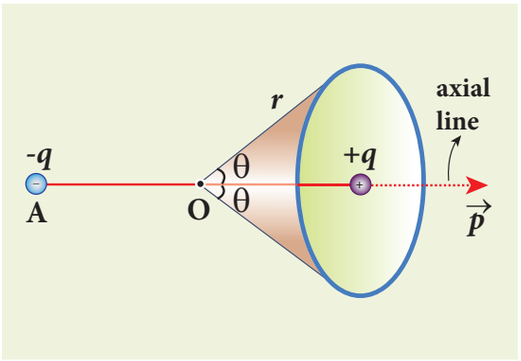


Figure 1.27 Dipole potential is axially symmetric

1.5.4 Equi-potential Surface

Consider a point charge q located at some point in space and an imaginary sphere of radius r is chosen by keeping the charge q at its center (Figure 1.28(a)). The electric potential at all points on the surface of the given sphere is the same. Such a surface is called an equipotential surface.

An equipotential surface is a surface on which all the points are at the same potential.

For a point charge the equipotential surfaces are concentric spherical surfaces as shown in Figure 1.28(b). Each spherical surface is an equipotential surface but the value of the